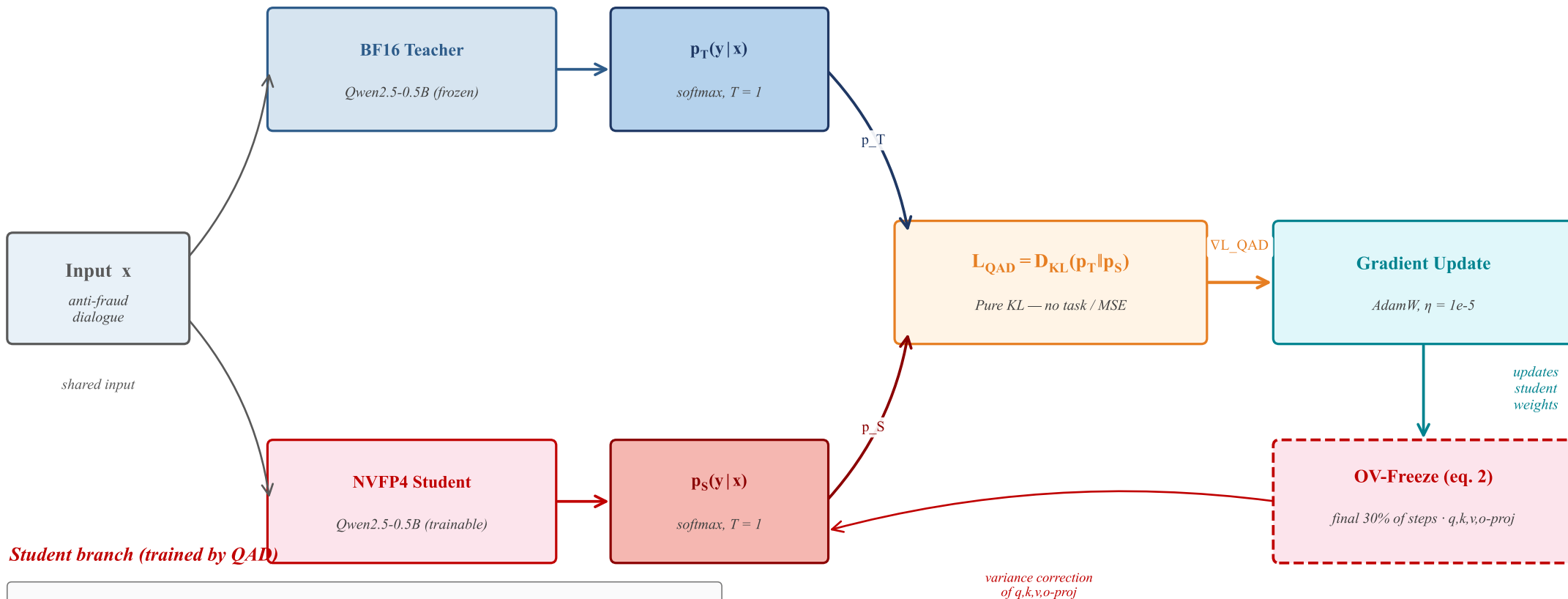


QAD Training Pipeline: Pure KL Distillation with Same-Source Teacher

Teacher branch



• 0.5B tokens (~1.7% of original SFT data) • $T = 1$ (no soft-label smoothing) • Same-source teacher • 99.1% recovery vs BF16

Fig. 9. The QAD pipeline. The BF16 teacher and the quantized student receive the same input; pure KL divergence flows back only to the student. OV-Freeze applies in the final 30% of training steps to the $q/k/v/o$ projections.